



SPRING MID SEMESTER EXAMINATION-2025

School of Computer Engineering
Kalinga Institute of Industrial Technology, Deemed to be University
Operating Systems
[CS20002]

Time: 1 1/2 Hours

Full Mark: 20

Answer Any four questions including question No.1 which is compulsory.

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

1. Answer all the questions.

[1 Mark X 5]

a) Three processes arrive at time zero with CPU bursts of 16, 20, and 10 milliseconds. If the scheduler has prior knowledge about the length of the CPU bursts, the minimum achievable average waiting time for these three processes in a non-preemptive scheduler (rounded to the nearest integer) is _____ milliseconds.

b) The following program consists of three concurrent processes and three binary semaphores. The semaphores are initialized as $S_0=1$, $S_1=0$, and $S_2=0$.

Process P0	Process P1	Process P2
while (True) {	wait (S1);	wait (S2);
wait (S0);	release(S0);	release(S0);
print('0')		
release(S1);		
release(S2); }		

How many times will the process P0 print '0'? Justify your answer.

a) At least twice b) Exactly thrice c) Exactly twice d) Exactly Once

c) The following C program is executed on a Unix/Linux system:

```
#include <unistd.h >
int main () {
    int i ;
    for (i=0; i<10; i++)
        if (i%2 == 0)
            fork ( ) ;
    return 0 ; }
```

The total number of child processes created will be _____ ? Give reason in support of your answer.

d) Consider a solution for the critical section problem. The given solution achieves mutual exclusion and bounded waiting only. Can the solution be considered as a valid one? Justify your answer.

e) A system is having 3 user processes P1, P2 and P3, where P1 requires 21 units of resource R, P2 requires 31 units of resource R, and P3 requires 41 units of resource R. The minimum number of units of R that ensures no deadlock is _____ ?

2. Answer the following questions: [3 + 2 = 5 Marks]

- (i): With the help of a diagram, show and explain the transition of a process from one state to another in the Five-State model.
- (ii): Differentiate between independent processes and cooperating processes. Explain race condition in process synchronization.

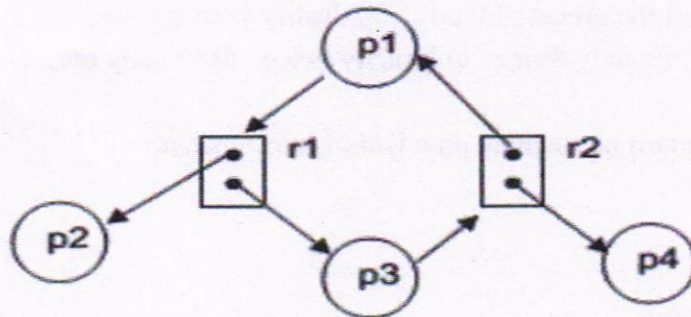
3. Answer the following questions: [3 + 2 = 5 Marks]

- (i): Explain Producer-Consumer problem and its solution using semaphores in details.
- (ii): Briefly explain the four necessary conditions for the deadlock to occur.

4. Consider the following table consisting of Process ID, Arrival Time (AT), Burst Time (BT), and Input/Output Burst (IOBT). The table describes that Process 1 arrives at 0 unit of time and requires 4 units of burst time, then goes for 5 units of I/O wait. After that, it returns to the Ready Queue to complete the remaining burst time, i.e., 3 units. Similarly, for Process 2 and Process 3. Compute the Average Turnaround Time and Average Waiting Time using the Round Robin Scheduling Algorithm with a time quantum of 2 units. [5 Marks]

PID	AT	BT1	IOBT	BT2
1	0	4	5	3
2	2	2	10	1
3	4	3	3	4

5. Explain various components of a Resource Allocation Graph (RAG). Consider the following RAG with 4 processes (p1, p2, p3, and p4) and 2 resources (r1 and r2).



Based on the given RAG, check whether the system is in safe state or not? If yes, give the safe sequence. [5 Marks]

*** Best of Luck ***